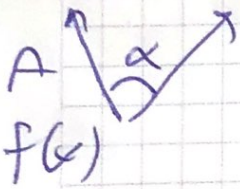
$f(x) = x \qquad g(x) = x^3 \qquad (-1, 1)$



$$\rightarrow \left(\frac{A}{|A|}\right)^T \cdot \left(\frac{B}{|B|}\right) = \cos\alpha$$

$$|f(x)| = \sqrt{\int_{-1}^{1} x^2 \, dx} = \sqrt{\frac{2}{3}}$$

$$|g(x)| = \sqrt{\int_{-1}^{1} x^6 \, dx} = \frac{1}{7} - -\frac{1}{7} = \sqrt{\frac{2}{7}}$$

$$\int_{-1}^{1} \frac{F(x)}{|F(x)|} \cdot \frac{g(x)}{|g(x)|} = \cos\alpha$$

sonuç
$\cos\alpha = \sqrt{\frac{21}{4} \cdot \frac{2}{5}}$
$\alpha = 23.5781$
(hesap makinesi)

$$\cos\alpha = \int_{-1}^{1} \frac{x}{\sqrt{\frac{2}{3}}} \cdot \frac{x^3}{\sqrt{\frac{2}{7}}} = \frac{21}{4} \int_{-1}^{1} x^4 = \frac{x^5}{5} \quad \frac{1}{5} - -\frac{1}{5}$$

$$= \sqrt{\frac{21}{4} \cdot \frac{2}{5}} \hookrightarrow \cos\alpha = \rightarrow \alpha = 23.57^\circ$$

$$\hookrightarrow \sqrt{\frac{21}{4} \cdot \frac{2}{5}} \rightarrow$$